

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**

**Product Identifier**

Perihalan Produk:

Product Description:

Cat No. :

**Larutan Boron**

**Boron solution 1000 ppm Aqueous solution**

J/8006/15, J/8006/05, J/8006/08

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use

Laboratory chemicals.

Uses advised against

No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
Selangor Darul Ehsan, Malaysia.  
Main line: +60 3-5525 7888

**Supplier**

E-mail address

Enquiry.my@thermofisher.com

Emergency Telephone Number

Tel: +03-5525 7888  
CHEMTREC Malaysia **1-800-815-308** (Malay)  
CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**

**Classification of the substance or mixture**

**Label Elements**

**Hazard Statements**

**Precautionary Statements**

**Prevention**

P201 - Obtain special instructions before use  
P202 - Do not handle until all safety precautions have been read and understood  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

**Response**

P308 + P313 - IF exposed or concerned: Get medical advice/attention

**Storage**

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P403 - Store in a well-ventilated place

## Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component                               | CAS No    | Weight % |
|-----------------------------------------|-----------|----------|
| Borates, tetra, sodium salts, anhydrous | 1330-43-4 | <0.3     |
| Water                                   | 7732-18-5 | >99.7    |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with plenty of water for at least 15 minutes. Get medical attention immediately if symptoms occur.

#### Ingestion

Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.

#### Inhalation

Remove to fresh air. Get medical attention immediately if symptoms occur.

#### Self-Protection of the First Aider

No special precautions required.

### Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

### Indication of any immediate medical attention and special treatment needed

#### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Use extinguishing measures that are appropriate to local circumstances and the surrounding environment.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

None reasonably foreseeable.

### Hazardous Combustion Products

Oxides of boron, Sodium oxides.

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## Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation. Use personal protective equipment as required.

### Environmental precautions

Should not be released into the environment. See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material. Sweep up and shovel into suitable containers for disposal.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Avoid contact with skin, eyes or clothing. Avoid ingestion and inhalation.

### Conditions for Safe Storage, Including any Incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place.

### Specific End Uses

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

| Component                               | Malaysia | ACGIH TLV                                             | OSHA PEL                            |
|-----------------------------------------|----------|-------------------------------------------------------|-------------------------------------|
| Borates, tetra, sodium salts, anhydrous |          | TWA: 2 mg/m <sup>3</sup><br>STEL: 6 mg/m <sup>3</sup> | (Vacated) TWA: 10 mg/m <sup>3</sup> |

| Component                               | European Union | The United Kingdom                                                | Germany |
|-----------------------------------------|----------------|-------------------------------------------------------------------|---------|
| Borates, tetra, sodium salts, anhydrous |                | STEL: 3 mg/m <sup>3</sup> 15 min<br>TWA: 1 mg/m <sup>3</sup> 8 hr |         |

### Exposure Controls

#### Engineering Measures

None under normal use conditions.

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles)

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## Hand Protection Skin and body protection

Protective gloves  
Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.  
(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection Recommended Filter type:

No protective equipment is needed under normal use conditions  
Particle filter

## Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

## Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### Information on basic physical and chemical properties

|                |                          |
|----------------|--------------------------|
| Appearance     | Colorless                |
| Physical State | Liquid                   |
| Odor           | No information available |
| Odor Threshold | No data available        |
| pH             | No information available |

|                     |                          |
|---------------------|--------------------------|
| Melting Point/Range | No data available        |
| Softening Point     | No data available        |
| Boiling Point/Range | No information available |
| Flash Point         | No information available |

Method - No information available

|                          |                   |
|--------------------------|-------------------|
| Evaporation Rate         | No data available |
| Flammability (solid,gas) | Not applicable    |
| Explosion Limits         | No data available |

Liquid

|                              |                          |
|------------------------------|--------------------------|
| Vapor Pressure               | No data available        |
| Vapor Density                | No data available        |
| Specific Gravity / Density   | No data available        |
| Bulk Density                 | Not applicable           |
| Water Solubility             | Miscible                 |
| Solubility in other solvents | No information available |

(Air = 1.0)

Liquid

### Partition Coefficient (n-octanol/water)

|                                         |         |
|-----------------------------------------|---------|
| Component                               | log Pow |
| Borates, tetra, sodium salts, anhydrous | -0.7570 |

|                           |                   |
|---------------------------|-------------------|
| Autoignition Temperature  | No data available |
| Decomposition Temperature | No data available |

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Viscosity No data available  
Explosive Properties No information available  
Oxidizing Properties No information available

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

None known, based on information available.

### Chemical Stability

Stable under normal conditions.

### Possibility of Hazardous Reactions

Hazardous Polymerization Hazardous polymerization does not occur.  
Hazardous Reactions None under normal processing.

### Conditions to Avoid

Incompatible products. Excess heat.

### Incompatible Materials

Strong oxidizing agents.

### Hazardous Decomposition Products

Oxides of boron. Sodium oxides.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

#### Product Information

##### (a) acute toxicity;

Oral Based on available data, the classification criteria are not met  
Dermal Based on available data, the classification criteria are not met  
Inhalation Based on available data, the classification criteria are not met

### Toxicology data for the components

| Component                               | LD50 Oral                 | LD50 Dermal                  | LC50 Inhalation                        |
|-----------------------------------------|---------------------------|------------------------------|----------------------------------------|
| Borates, tetra, sodium salts, anhydrous | LD50 = 2660 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rabbit ) | LC50 > 2 mg/m <sup>3</sup> ( Rat ) 4 h |
| Water                                   | -                         | -                            | -                                      |

(b) skin corrosion/irritation; No data available

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(c) serious eye damage/irritation; No data available

(d) respiratory or skin sensitization;

Respiratory No data available  
Skin No data available

(e) germ cell mutagenicity; No data available

(f) carcinogenicity; No data available

There are no known carcinogenic chemicals in this product

(g) reproductive toxicity; No data available

(h) STOT-single exposure; No data available

(i) STOT-repeated exposure; No data available

Target Organs None known.

(j) aspiration hazard; No data available

Symptoms / effects, both acute and delayed No information available.

**Endocrine Disrupting Properties** Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### Ecotoxicity effects

| Component                               | Freshwater Fish                            | Water Flea                                        | Freshwater Algae                                                                                                                      | Microtox |
|-----------------------------------------|--------------------------------------------|---------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|----------|
| Borates, tetra, sodium salts, anhydrous | LC50: = 340 mg/L, 96h<br>(Limanda limanda) | LC50: 1085 - 1402<br>mg/L, 48h (Daphnia<br>magna) | EC50: 2.6 - 21.8 mg/L,<br>96h static<br>(Pseudokirchneriella<br>subcapitata)<br>EC50: = 158 mg/L, 96h<br>(Desmodesmus<br>subspicatus) |          |

### Persistence and degradability

**Persistence** Miscible with water, Persistence is unlikely, based on information available.

**Bioaccumulative potential** Bioaccumulation is unlikely

| Component                               | log Pow | Bioconcentration factor (BCF) |
|-----------------------------------------|---------|-------------------------------|
| Borates, tetra, sodium salts, anhydrous | -0.7570 | No data available             |

### Mobility in soil

The product is water soluble, and may spread in water systems. . Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

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Other adverse effects No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

### Waste treatment methods

**Waste from Residues/Unused Products**

Dispose of in accordance with local regulations

### Contaminated Packaging

Empty containers should be taken to an approved waste handling site for recycling or disposal

## SECTION 14: TRANSPORT INFORMATION

### IMDG/IMO

Not regulated

### Road and Rail Transport

Not regulated

### IATA

Not regulated

### Special Precautions for User

No special precautions required

## SECTION 15: REGULATORY INFORMATION

### Safety, health and environmental regulations/legislation specific for the substance or mixture

### International Inventories

X = listed

| Component                               | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------------------------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Borates, tetra, sodium salts, anhydrous | 215-540-4 | X    | X   | X     | X    | X    | X     | X    | KE-12384 |
| Water                                   | 231-791-2 | X    | X   | X     | X    |      | X     | X    | KE-35400 |

### National Regulations

### Persistent Organic Pollutant

This product does not contain any known or suspected substance

### Ozone Depletion Potential

This product does not contain any known or suspected substance

| Component                               | Persistent Organic Pollutant | Ozone Depletion Potential | Pesticides Act 1974 |
|-----------------------------------------|------------------------------|---------------------------|---------------------|
| Borates, tetra, sodium salts, anhydrous |                              |                           | X                   |

## SECTION 16: OTHER INFORMATION

### Legend

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**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

## Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

Revision Date

23-Mar-2025

Revision Summary

SDS sections updated.

**In accordance with local and national regulations: Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**